1. Random Initialization

Random initialization is the method we discussed in lecture. One common approach, called Forgy's method [1], selects (K) random data points and uses them as the initial centroids. Another approach, called Random Partition [2], randomly assigns each data point to a cluster and calculates the mean of each cluster to get the starting centroids.

The main advantage of both methods is that they are simple to implement and have low initialization costs. However, the results can vary depending on the random choices. For example, two centroids might be placed in the same natural cluster, leaving another cluster without a good starting centroid. This can lead to poor final results. Random Partition can also produce poor starting centroids because the randomly assigned groups may have similar means. One way to improve the results is to run K-means multiple times with different initializations and keep the run with the lowest objective function value (J). However, this also increases the total runtime. Peña et al. [3] compared several initialization methods and found that the choice of initialization can have a significant effect on clustering performance.

2. Maximin (Farthest-First)

Maximin starts by selecting a data point as the first centroid. It then repeatedly selects the point that is farthest from its nearest existing centroid until (K) centroids have been chosen [4].

The main advantage of this method is that it spreads the initial centroids across the dataset. This can help avoid placing all the centroids in the same region. It is also relatively straightforward to implement. However, the method is sensitive to outliers. Since it always selects the farthest point, an outlier may be chosen as a centroid even if it does not represent a meaningful cluster. This can lead to worse clustering results. Maximin can be useful when clusters are well separated and the data does not contain many outliers, but it may not work as well with noisy data.

3. K-means++

K-means++ was introduced by Arthur and Vassilvitskii [5] to improve the selection of initial centroids. It starts by choosing one data point randomly. For each remaining point, it calculates the squared distance to the nearest selected centroid. The next centroid is then selected randomly, with points farther from the existing centroids having a higher probability of being chosen. This process continues until (K) centroids have been selected.

Compared with basic random initialization, K-means++ generally provides a better starting point because it encourages the centroids to cover different regions of the dataset. The authors also proved that its expected clustering objective has an (O(\log K)) approximation guarantee. However, initialization requires additional distance calculations, which can be expensive for large datasets. It is also randomized, so different runs can still produce different results. Although K-means++ can select outliers, it does not always choose the farthest point, as Maximin does.

Conclusion

Based on these three methods, K-means++ is a good choice for most applications because it usually provides better initial centroids than basic random initialization without always selecting the farthest points.

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